Company Name: AAIL - Applied Artificial Intelligence Laboratory

Company Profile:

Transforming Education Through Advanced AI

AAIL is a pioneering technology firm dedicated to revolutionizing the education sector through the power of Artificial Intelligence. With a proven track record of success, we specialize in developing scalable, user-friendly solutions that bridge the gap between traditional teaching methods and the evolving needs of the modern digital environment.

Our Reach & Impact

Our commitment to excellence is reflected in the widespread adoption of our flagship products—Sayo, Penso, Makwan, and more. Currently, AAIL serves over 150 schools across Hong Kong and the Greater Bay Area. Our platforms are widely recognized for their ability to deliver AI-driven personalized learning pathways and real-time interactive features, empowering both educators and students to achieve more.

Core Technology & Expertise

At the heart of AAIL is a team of experts with deep capabilities in Large Language Models (LLMs), Natural Language Processing (NLP), and Computer Vision. Unlike generic solutions, we pride ourselves on our ability to perform in-house training and fine-tuning of models. This technical rigor allows us to create truly adaptive learning systems tailored to the unique requirements of educational institutions.

Digital Transformation for Enterprise

Beyond the classroom, AAIL is a trusted partner for leading publishers and telecom operators. We drive digital transformation by "super-charging" traditional business models. By integrating AI with traditional formats, such as printed media, we unleash new potential, turning static content into interactive, adaptive experiences that engage users on a deeper level.

Industry Leadership

AAIL is more than a service provider; we are a thought leader in the EdTech space. We maintain a strong presence within the industry through frequent collaborations with the universities, EdCity, Education Bureau Hong Kong and leading IT associations. As active speakers and organizers of AI-related competitions, we are committed to fostering innovation and shaping the future of education technology.

Project 1:

Next-gen multimodal AI that sees, hears, reads, and gives human-like feedback

Introduction:

Grading shouldn't be limited to text. Sayo Grader is a next-gen AI platform designed to assess the entire student experience. Leveraging Multimodal LLMs, we are building a system that can see, hear, and read to provide comprehensive feedback, bridging the gap between human intuition and AI efficiency.

What this project does:

A unified platform that accepts Visual, Audio, and Text inputs. It scans handwriting (OCR), listens to oral presentations, and reviews essays. The system outputs scores plus multimodal feedback, including generated voice notes for encouragement and visual annotations on exam papers.

Key areas of development:

1. AI & ML Pipeline: OCR, Speech-to-Text (Whisper), and prompt engineering for LLMs.

2. Full-Stack Engineering: Building a React frontend for media uploads and a Python/FastAPI backend.

3. UX Design: Creating a "Human-in-the-loop" dashboard for teachers.

Deliverables:

1. Functional MVP: A web platform capable of grading images, audio, and text.

2. Smart Algorithm: A customizable rubric engine with adjustable strictness.

3. Demo Showcase: A polished real-time grading presentation.

4. API Documentation for system architecture.

What students can learn:

GenAI Application: Hands-on experience with Multimodal AI. Modern Tech Stack: Proficiency in Python, Vector Databases, and LLM APIs. Product Development: Transforming a concept into a deployed prototype. Data Handling: Solving real-world issues like bad handwriting or noise.

Impact:

Tech Frontier: Pioneering "sensory" AI in EdTech beyond text-checkers. HK Education: Drastically reducing grading burdens for teachers. Student Benefit: Providing instant, personalized feedback in voice or visual formats that feel human.

Conclusion:

Sayo Grader is more than a coding project; it is an attempt to redefine assessment. We are looking for interns ready to build a tool that directly shapes the future of the smart classroom.

Project 2:

AI-Driven Market Intelligence & Lead Scoring System

Introduction:

In the fragmented EdTech landscape of Hong Kong, identifying schools ready for digital transformation is a manual and time-consuming process. Our Business Development team currently relies on scattered public data to find prospects. This project aims to automate market research by building a data pipeline that aggregates school profiles and utilizes a "Lead Scoring" engine to rank schools based on their likelihood to adopt AI solutions.

What this project does:

- Automates Data Collection: Scrapes and aggregates public data from the EDB, school websites, and annual reports.
- Analyzes Readiness: Uses NLP to scan school mission statements for tech-forward keywords (e.g. ""STEM"", ""BYOD"")
- Scores Leads: Assigns a ""Sales Readiness Score"" to every school based on funding, infrastructure, and curriculumn focus
- Visualize Markets: Displays schools on an interactive map to help the BD team prioritize outreach.

Key areas of development:

1. Web Scraping & Data Engineering: Building robust crawlers to handle various web structures.

2. NLP: Extracting structured insights from unstructured text documents

3. Algorithm Design: Develeoping a weighting logic to quantify qualitative school data.

4. Data Visulization: Creating a user-friendly dashboard

Deliverables:

Data Pipeline Scripts: Python code for scraping, cleaning, and updating school data.

Scoreing Model: The algorithm and logic used to rank schools.

Interactive Dashboard: A web-based map interface for the internal team to filter and view school data.

What students can learn:

handson experience on data science and business intelligence. learn how to handle "dirty" real-world data, bypass scraping anti-bot measures, apply NLP for business insights, and build full-stack data applications that solve immediate commercial problems.

Impact:

Tech Frontier: Applying data science to sales strategy (Revenue Operations), a high-growth field.

HK Education Industry: Promotes efficient resource allocation by ensuring schools actively seeking innovation are identified and supported first, helping bridge the digital divide faster.

Conclusion:

This project addresses a critical operational bottleneck. By replacing manual research with automation, it not only optimizes our business strategy but gives students the rare opportunity to build a market intelligence engine that directly influences how EdTech reaches Hong Kong classrooms.

Project 3:

Retrieval Augmented Generation System for Internal Knowledge Management

Introduction:

As our EdTech product suite grows, so does our technical documentation. A major challenge is the "knowledge silo" where non-technical staff struggle to find accurate answers to client questions quickly. This project aims to build an internal Retrieval-Augmented Generation (RAG) Chatbot. By ingesting our internal knowledge base, this AI assistant will allow staff to query complex product questions in natural language and receive instant, cited answers.

What this project does:

- Centralizes knowledge: Indexes disparate documents (PDFs, Google Docs, Notion) into a vector database.
- Answers Queries: Provides a chat interface for staff to ask technical questions
- Ensures Accuracy: Uses LLMs to synthesize answers strictly from internal data to prevent hallucinations.
- Cites Sources: Provides direct links to the source document for verification.

Key areas of development:

1. Generative AI & LLM: Implementing LangChain or LlamaIndex to orchestrate model behavior.

2. Vector Search: Managing vector databases (3.g. Pinecone, Milvus) for semantic information retrieval.

3. Backend Development: Building APIs to handle document ingestion and chat streams.

4. Frontend Interface: Designing a clean, intuitive chat UI for internal employees. (**can open for external use)

Deliverables:

Functional RAG Pipeline: Backend system for processing documents and handling queries.

Chat Application: A web interface or Slack bot integration for user interaction.

Adming Dashboard: A simple UI for uploading and managing the internal knowledge base.

Deployment Guide: Instructions for hosting the model securely with data privacy protocols.

What students can learn:

A deep dive into Generative AI Application Development, the most in-demand skill in tech today. Students will learn Prompt Engineering, Vector Embeddings, and how to build "Production-Grade" AI that balances accuracy, speed, and data privacy.

Impact:

Tech Frontier: Practical implementation of RAG, currently the standard for enterprise AI adoption.

HK Education Industry: Significantly speeds up technical support for schools. By automating internal knowledge sharing, our bd team can spend less time searching/ communicating with engineer for answers and more time supporting teachers and students.

Conclusion:

The GenAI Knowledge Assistant represents the intersection of operational efficiency and advanced AI. It solves a real communication problem within a fast-paced startup. Students will leave this project with a portfolio piece that demonstrates their ability to build secure, functional GenAI tools for enterprise use.

Project 4:

AI 手写识别技术与教育产品应用

Introduction:

本项目旨在让学生探索并实作「中英文手写 OCR（Optical Character Recognition）」技术，并最终为 AI 教育产品提供可直接使用的辨识模块、测试数据与技术建议。学生将不仅进行研究，更会产出能融入产品开发流程的实用成果。

What this project does:

学生将建立一套可辨识「中英文手写文字」的 OCR 解决方案，并针对不同模型与技术进行测试与评估。项目将产出可整合到教育产品中的原型模块，协助处理学生手写答案的自动批改、数据化与分析功能。

项目包括模型测试、数据集建立、前处理/后处理策略，以及呈现可供产品团队使用的技术成果。

Key areas of development:

1. 中英文手写 OCR 技术探索与整合

在多种技术中选出可应用于 AI 教育产品的方案（例如 Tesseract、PaddleOCR、Vision Models）。

2. 影像前处理流程设计

提高手写识别准确度（如影像校正、去噪、二值化、切割字词）。

3. 数据集建立与训练/微调

建立与实际产品场景相符的手写数据集，并可尝试简单模型微调。

4. 效能 Benchmark 建立与测试

包含准确率、字词错误率（CER/WER）、速度、对不同字体的容错能力。

5. 产品原型 Demo 开发

建立可供产品使用的 OCR 原型流程（如：输入图片 → 辨识 → 产出文本）。

Deliverables:

1. OCR 技术比较与效能评估报告，可直接提交给工程与产品团队。

2. 完整的中英文手写数据集。

3. 一套可整合至产品的 OCR 原型 Demo 或程序流程。

4. 提升辨识准确度的前后处理模块。

5. Benchmark 测试文件与未来改善建议。

6. 项目成果简报，包含研究、实作与产品应用方向。

What students can learn:

学生可学到 AI 模型原理、OCR 技术应用、数据集建立与图像处理、模型测试与效能比较、如何设计能落地的 AI 技术流程、技术文件撰写、产品思维与跨部门协作能力。这些技能均能与真实科技产品开发接轨。

Impact:

1. 技术前沿：学生将接触最新的 OCR 与 AI 模型技术，理解语言模型与传统 OCR 结合的可能性。

2. 香港教育界：可推动无纸化学习、自动批改、学习历程纪录等教育科技应用。

3. 产品影响：成果可直接用于公司 AI 教育产品，例如自动批改手写答案、手写字迹辨识、学生作答数据化、作业流程自动化等。

4. 学生影响：学生能掌握真实 AI 技术落地能力，提升未来于科技、数据与产品领域的竞争力。

Conclusion:

本项目将协助学生从探索到实作，完整掌握中英文手写 OCR 技术，并产出可直接应用于 AI 教育产品的技术成果。学生不只学习技术，更学习如何将 AI 技术落地到真实产品中，对公司与学生自身均具有长远价值。

Project 5:

AI-Driven Crowdsourced Language Acquisition Platform

Introduction:

In the current landscape of English language education, learners often face a disconnect between rote vocabulary memorization and real-world application. While students encounter new words daily through various media—such as video games, movies, and social media—they lack a centralized tool to capture these instances and convert them into meaningful learning experiences. The project addresses this gap by utilizing Generative AI (GenAI) to transform a user's passive vocabulary encounters into active, contextualized learning materials. This project bridges the divide between entertainment and education, ensuring that learning is relevant, personalized, and engaging.

What this project does:

A multi-platform ecosystem (Mobile and Web App) designed to enhance English skills through two distinct, interconnected modes:

- **The Contextual Dictionary** (Capture Mode):
  Users input unknown vocabulary words they encounter in their daily lives. Crucially, users also input the source or context of the word (e.g., Gaming, Cinema, Literature, News).
- **AI Integration**: The system define the word by calling dictionary, while categorizing the input into genres.
- **The Generative Learning Arena** (Practice Mode):
  The system aggregates the ""unknown words"" collected from the user base (or the individual user's history).
- **AI Content Generation**: Using Large Language Models (LLMs), the app dynamically generates unique reading passages that incorporate these specific target words.

Example: If users recently learned the words ""potion,"" ""inventory,"" and ""quest,"" the AI generates a short story set in a fantasy world containing these words, followed by comprehension questions

Key areas of development:

1. Cross-Platform Development: Utilizing frameworks like React Native for mobile, and Next.js for the web portal, ensuring a unified user experience.

2. Generative AI & NLP: Integration with LLM APIs (e.g., OpenAI GPT, Gemini) to handle genre classification and the dynamic generation of educational content.

3. Backend Architecture: Building a robust server to manage user data, the global vocabulary bank, and API orchestration.

4. Cloud Database Management: Designing a SQL schema to track user progress, word retention rates, and genre preferences.

Deliverables:

1. Multi-Platform Application MVP: A fully functional dictionary mobile app (iOS/Android) and a responsive web application.

2. AI Engine Module: A backend service capable of classifying vocabulary context and generating coherent, genre-specific stories and quizzes.

3. User Dashboard: A tracking interface for users to visualize their ""collected words,"" learning streaks, and mastery levels.

4. Admin Panel: For educators or admins to monitor the ""trending words"" being searched by users.

Project Documentation: Comprehensive technical report, API documentation, and a user manual.

What students can learn:

1. Full-Stack Engineering: Gaining proficiency in connecting front-end interfaces with complex back-end logic and databases.

2. Prompt Engineering & AI Integration: Learning how to effectively instruct LLMs to output structured educational data (JSON) and creative content simultaneously.

3. State Management: Handling real-time data synchronization between the web and mobile platforms.

4. UX/UI Design for EdTech: Understanding how to design interfaces that maximize learner retention and minimize cognitive load.

Impact:

"Tech Frontier: This project aligns with the current surge in Adaptive Learning and Generative AI. It moves beyond static content (standard textbooks) to ""Just-in-Time"" content generation, representing the future of personalized education.

HK Education Industry: In a bilingual city like Hong Kong, students often struggle to apply ""textbook English"" to the Western media they consume (Netflix, YouTube, Gaming). This tool validates their interests (e.g., learning English through gaming) and converts their leisure time into productive study, potentially increasing English proficiency across the student body.

Conclusion:

This project is not just a dictionary; it is a dynamic content engine that evolves with the user. By capturing vocabulary at the moment of interest and recycling it into custom-made learning materials, the project proposes a sustainable loop of acquisition and practice. It challenges students to master cutting-edge AI technologies while solving a fundamental human problem: making language learning sticky, relevant, and fun.

Project 6:

智能交互式动画学习内容设计系统

Introduction:

本项目是一个创新的教育工具，结合了 AI 生成的播客音讯与同步的动画，旨在提供交互式的学习体验。透过 AI 技术，系统能自动将复杂概念/ 教科书转化为生动的可视化内容和讲解音讯，让学习变得更加直观有趣。

What this project does:

- 接收用户输入的概念描述和难度等级
- 可选择性上传 PDF 教材作为参考数据
- 自动生成详细的教学脚本
- 创建动画来可视化概念
- 生成同步的语音导览解说
- 输出完整的交互式教学页面

Key areas of development:

- 可视化设计: 如何将抽象概念转化为动态图像,设计有意义的动画流程
- 服务架构: 建立稳定的服务端点,处理使用者请求与响应
- 内容生成: 整合 AI 能力来创造教学内容,理解如何下达有效指令
- 多媒体整合: 协调视觉与听觉元素,创造流畅的学习体验
- 项目规划: 从零开始设计完整的应用程序架构,处理错误与资源管理

Deliverables:

- 功能完整的 Web 应用程序
- 完善的 API 文件说明
- 范例教学动画展示

- 技术文件与使用指南
- 项目报告(包含设计理念、实作过程、测试结果)

What students can learn:

- 理解 AI 的实际应用场景
- 掌握前后端分离的开发模式
- 学习处理多媒体内容的技术
- 培养系统整合与除错能力
- 体验完整的软件开发生命周期
- 提升问题分析与解决能力
- 了解如何将 AI 技术应用于教育领域

Impact:

- 教育科技: 为教师提供创新的教学内容制作工具,降低数字教材制作门坎
- 技术人才培育: 培养学生整合多种技术的实战能力,为进入科技产业做准备
- 开源贡献: 可作为教育开源项目,供其他开发者学习和改进

Conclusion:

这个项目展示了 AI 技术如何赋能教育创新,学生将透过实际开发,深入理解现代 Web 应用的完整开发流程。从需求分析、系统设计、功能实现到测试优化,学生将获得宝贵的实战经验。更重要的是,这个项目具有实际应用价值,可以真正帮助教师创造更好的教学内容,体现了技术服务教育的理念。